



The *building*

All life is based on chemistry. There are 92 natural elements on Earth. These are the substances from which all other things

CRUCIAL CARBON

Carbon is the most important element for life on Earth.

It has a unique ability to form molecules of different shapes, especially long chains and hexagonal rings. Molecules that contain carbon are called organic compounds. There are four types of carbon compound that are crucial to all living things: carbohydrates, lipids, proteins, and nucleic acids.

There's no getting away from me. I'm totally essential for all living things!

In the human body, carbon is the *second most abundant element* after *oxygen*.

Carbohydrates are made of carbon rings with hydrogen and oxygen atoms attached. The simplest types have just one carbon ring and these include sugars, which provide energy. Sugars can be found in many foods, such as honey, fruits, and berries. More complex carbohydrates form long and branching chains of rings and include starch and cellulose, which are found in plants.

Lipids are greasy or waxy substances that include fats and oils. They are made of long chains of carbon and hydrogen atoms. Lipids are needed to form cell membranes (the outer surfaces of cells) and are a useful way to store energy. The human body can make some lipids on its own, but has to get others from the things we eat, such as animal fat, butter, and cooking oil.

Proteins are vital molecules in any organism. They are used for everything from building cells (the simplest living part of an organism) to speeding up reactions and transporting other molecules. Proteins are large, and are made up of smaller units called amino acids. Although there are more than 200 different amino acids, the proteins in most organisms are made from just 20.

Nucleic acids carry the instructions for making proteins and, therefore, all living things. They contain information that controls how a cell functions and reproduces. There are two types: RNA (ribonucleic acid) and DNA (deoxyribonucleic acid). DNA is the most important molecule in the body.

